

Why did you choose to work with a 28-nanometre technology node?

Sharan: Internally, we refer to the 28-nanometre process node as the ‘Goldilocks node’ because it offers sufficient speed, incorporates power-saving features in the node’s design, and is relatively cost-effective. It is the last process node where traditional planar transistors are used before transitioning to FinFET transistors, which are more complex and expensive, particularly in the case of 16-nanometre and lower nodes. Developing for FinFET nodes presents greater challenges and costs, making 28 nanometres an ideal choice that provides the desired performance, power efficiency, and manageability of process technology without breaking the bank. Hence, the fitting nickname ‘Goldilocks.’

Shashwath: Though it was not a primary factor in our decision-making, the 28-nanometre process node does not pose export control issues from the US. Access to this node is freely granted. In contrast, for the latest process nodes, a separate licence is required, and the associated paperwork could have caused significant delays, possibly extending to around six months or even longer. So, beyond its technical advantages, the 28-nanometre node also offers streamlined access, aligning perfectly with our needs for the first year of development.

Can you give me an example of where 28-nanometre nodes are used in real life?

Sharan: Xilinx FPGAs, such as Artix 100 and DS 200, are typically fabricated at 28 nanometres. The first-generation Vertex 7 series was produced at 28 nanometres. The very first version of the Apple Watch was manufactured using a 28-nanometre process. Many

automotive chips are manufactured using the 28-nanometre process.

How does one choose a process node for a particular chip?

Sharan: The choice of process node varies depending on the application and use case. It also depends on when that chip was initially designed. Several smartphone chips from around 4 to 6 years ago commonly used the 28-nanometre node. Until about two years ago, most entry-level smartphones, falling within the price range of 6000 to 15,000 rupees, commonly used chips based on the 28-nanometre node. These smartphones typically feature processors from MediaTek. However, in 2020 and 2021, there was a shift towards the 16-nanometre process, which has become the standard even for lower-end smartphones. This transition marks a notable evolution from this category’s previously widespread use of the 28-nanometre process.

What is the reason for using higher technology nodes in the automotive sector?

Shashwath: In the automotive sector, this choice is driven by considerations related to thermal management. In contrast to FinFET transistors, where the fins are closely packed and radiate heat outward, 28-nanometre chips have a different configuration that allows them to dissipate heat more effectively. The orientation of the fins plays a crucial role in how heat is managed. FinFETs, with their vertical fin design, tend to radiate heat upwards, potentially affecting adjacent components. Thermal dissipation can indeed be a challenge in the automotive industry. The ambient temperature in which these chips operate is already relatively high. As a result, many controller-class chips inside electronic control units (ECUs) in

automobiles are manufactured using the 28-nanometre process or higher.

Why are smaller technology nodes garnering so much attention?

Sharan: In the realm of smartphones, only so much physical space is available. Yet, as consumers, we demand vast features within these compact devices. We want to stream Netflix in 4K, play games seamlessly without lag, and achieve a gaming experience akin to what we get on a PC with a dedicated Nvidia GPU, all within this limited real estate. The challenge becomes squeezing as much computational power as possible into a small area, hence the competitive push towards chips like the 10, 7, 5, 4, nanometres, or other variants that aim to maximise performance within constrained dimensions.

How do smaller technology nodes impact the design of different categories of electronic devices?

Shashwath: As transistors shrink, their power consumption decreases. This power efficiency is vital in devices like smartphones, where extended battery life is a top priority. However, in certain contexts, such as electric vehicles (EVs), where the power consumption ranges from a few watts to several kilowatts due to the motor’s requirements, the impact of shaving off a few dozen watts is relatively insignificant. Therefore, design practices may differ for these diverse use cases, and what’s essential for a smartphone may not apply to an EV.

The second area where scaling down is significant is in servers for data centres and cloud computing. Despite the seemingly ample space, data centres face heat dissipation and spatial optimisation challenges. The proximity of racks can lead to heat transfer, making efficient cooling crucial, especially when